

Boise State University
Department of Electrical and Computer Engineering
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Design Project

Modeling of a Cart and Track System

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1. OBJECTIVES

Control systems can be evaluated by three different methods: equation-based analysis, model-based analysis, and estimation using real data. The cart and track system has a transfer function with unknown a and b parameters. Equation-based analysis calculates these parameters by using a known relationship between the parameters and characteristics of the cart and track system. The model-based analysis uses a block diagram of the system using Simulink, where the unknown parameters are found by using the backward difference method and relating velocity to the parameters. Finding the unknown values with real data is like the model-based approach, where real data is used instead of simulated data. This report goes into the different methods to come up with these parameters.

2. Transfer Function

Fig. 1 is the system that is being modeled. It is a cart on an inclined hill, at angle Φ . The cart has wheels that have front and back wheels with different functions. The encoder wheel measures the position of the cart using an encoder. The powered wheel is responsible for moving the cart and is the input of the system. The output of the system is the position of the cart.

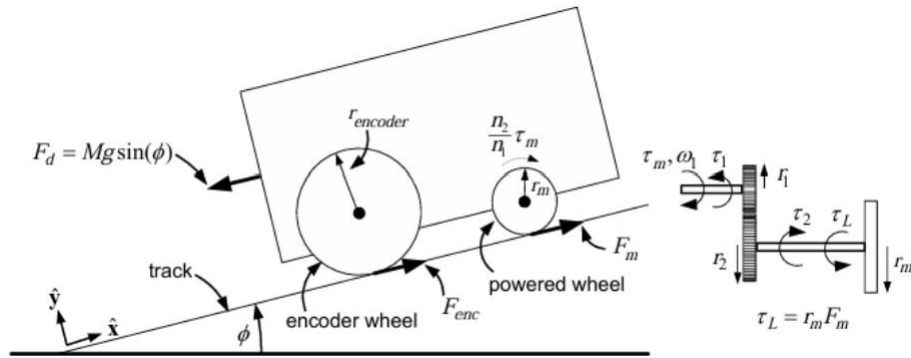


Figure 1: Cart on a inclined floor, powered wheel gears

Fig. 2 is the simplified transfer function of the cart and track system shown in Fig. 1. The input, $V_a(s)$, is the input voltage from the powered wheel. The disturbance, $K_D F_d(s)$, is a gravitational force that is reliant on Φ , the incline angle. $G(s)$ translates how the input and disturbance are related to $X(s)$. The parameters in $G(s)$, a and b , affect the response of the system. The parameter a affects the stability and response time while b is the system gain. $X(s)$ is the output of the system, referring to the position of the cart.

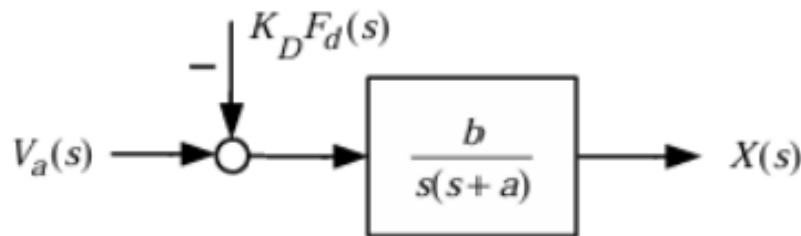


Figure 2: Simplified transfer function of the cart and track system

Fig. 3 is the Simulink model of the cart and track system. The central block represents the block diagram shown in Fig. 2. The section on the right of Fig. 3 is the encoder model, used to simulate position while using an encoder. There are two methods used to calculate the velocity of the cart, the differentiation and backward difference method. For future calculations, the velocity value used will come from the backward difference method. The bottom of Fig. 3 is responsible for collecting data and storing it into a data file.

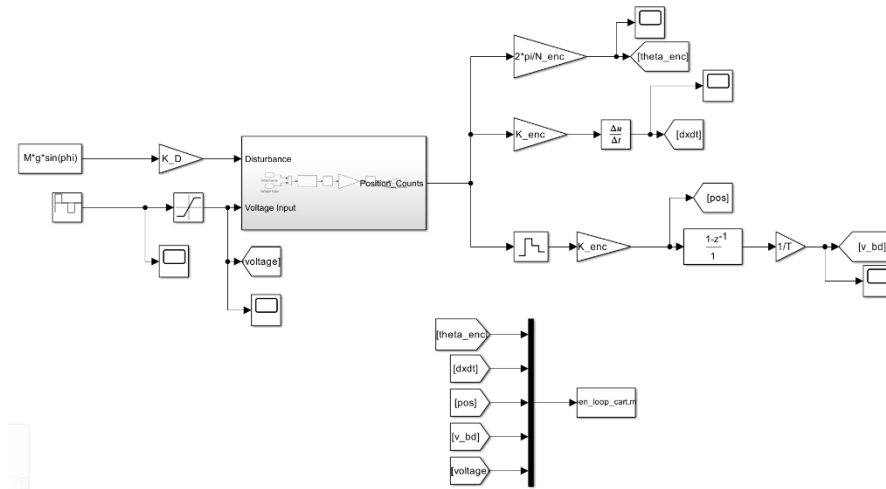


Figure 3: Simulink model of the cart and track system

2.2 Required Graphs

Fig. 4 is a velocity and time graph, where the velocity is obtained through the backward difference method. During this time, the input voltage has changed every second by unit steps. For a constant voltage of 2V from 0 to 1 second, the steady state velocity is 0.455 m/s.

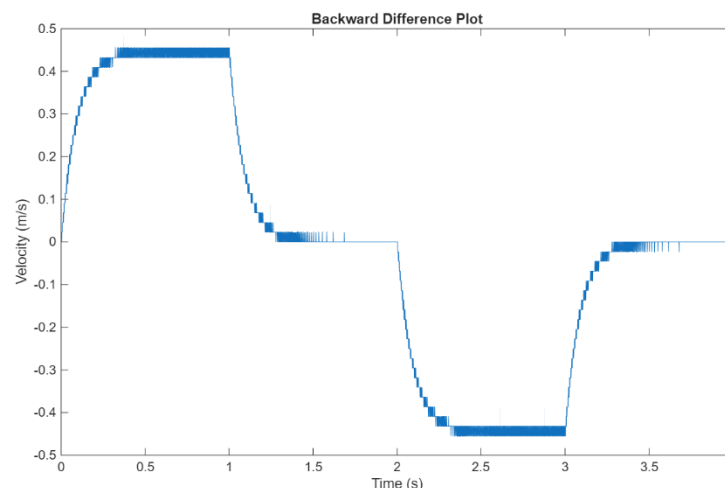


Figure 4: Backward difference plot with the simulation-based analysis

Fig. 5 is a position and time plot from the simulation-based analysis. The position of the cart increase, is constant, then decreases over time. This is also the integration of Fig. 4.

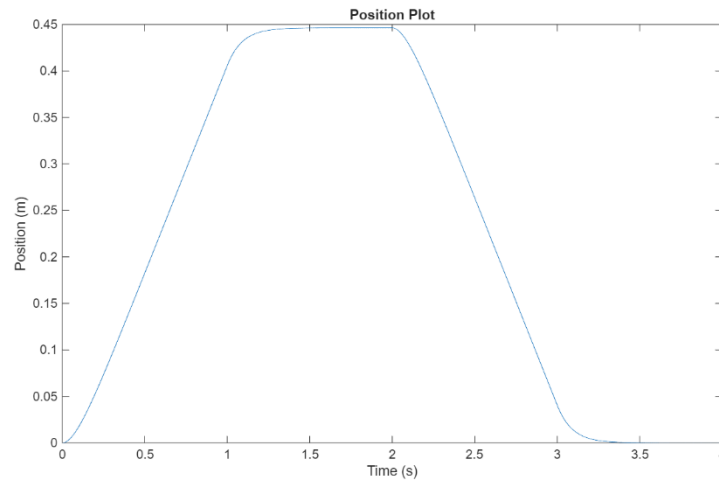


Figure 5: Position plot with the simulation-based analysis

Fig. 6 is the error of the backward difference method. It is expected that this difference is within $\pm$ of K_{enc}/T , or ± 22.75 mm. Visually, the error from Fig. 6 does not exceed 22.75 mm and is within the expected error range.

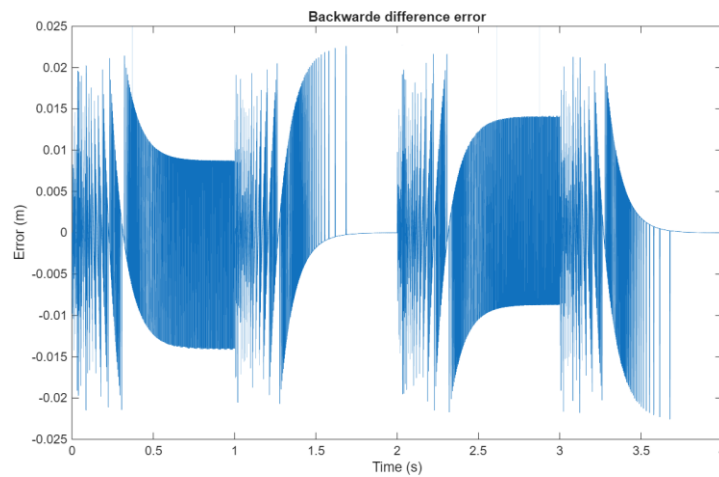


Figure 6: Backward difference error plot of simulated-based analysis

3. System Parameters

Table 1 is the parameter values used for the cart and track system. For the equation-based analysis, these parameter values were used to calculate for a and b.

Table 1: Parameter values for the cart and track system

Parameter	Value
KT	$7.67 \times 10^{-3} \text{ Nm/A}$
R	2.6 Ohms
L	0.18 mH
MC	0.57 kg
MW	0.37 kg
rM	$6.35 \times 10^{-3} \text{ m}$
J1	$3.9 \times 10^{-7} \text{ kgm}^2$
J2	0
Jenc	0
f1	0
f2	0
fenc	0
Rg	3.71
g	9.8 m/sec^2
Nenc	4096 counts / rev
Kenc	$2.275 \times 10^{-5} \text{ meters/count}$
Vmax	5V
T	0.001 sec

The equations used for calculating a and b are:

$$a = \frac{\frac{Rf}{r_m^2} + K_b K_T \left(\frac{n_2}{n_1} * \frac{1}{r_m} \right)^2}{\frac{RJ}{r_m^2} + RM} \quad (1)$$

and

$$b = \frac{\frac{K_T n_2}{n_1} 1/r_m}{\frac{RJ}{r_m^2} + RM} \quad (2)$$

3.1 Estimated System Parameters

Table 2 is the solved a and b parameters. The equation-based values come from equations 1 and 2, using the parameters in Table 1. The values for a and b using the equation-based analysis are 10.98 and 2.45 respectively.

The estimation based using a simulation tool is fairly accurate and yield similar results to the equation-based analysis. The simulation uses the block diagram from Fig. 3 and derive a and b by using the response of the system. $a = 1/T_m$ where T_m is the time where the velocity is at 0.632 of its steady state value. For the b parameter, it is calculated by ratio of the output and input, multiplied by a. The values of a and b are 10.87 and 2.47 respectively. This is within $\pm 1\%$ of the equation-based values.

The analysis using real data is done similarly to the simulation-based method. Instead of using data obtained from the simulation, data is measured on the actual system and used for the calculations of a and b . The values of a and b using real data are 9.71 and 1.88 respectively. The values do vary from the equation-based and simulation-based values. A big reason for this discrepancy is that the model did not consider friction.

Table 2: System Parameters

<i>Parameter</i>	<i>Equation</i>	<i>Estimation based on model</i>	<i>Estimation based on real data</i>
a	10.98	10.87	9.71
b	2.45	2.47	1.88

4. CONCLUSION

It is important to model systems accurately, so the response of the system is more accurately predicted. This makes the control of the system more precise. There are three common ways of modeling a system: equation-based analysis, simulation-based analysis, and the use of real measured data on the system. All these methods are valid ways to model a system, but they do have their own benefits and disadvantages. Generally, equation-based analysis can easily be automated and is quicker to calculate. Using real measured data to model a system is more time consuming, however it is the most accurate when modeling a system.

APPENDIX (A)

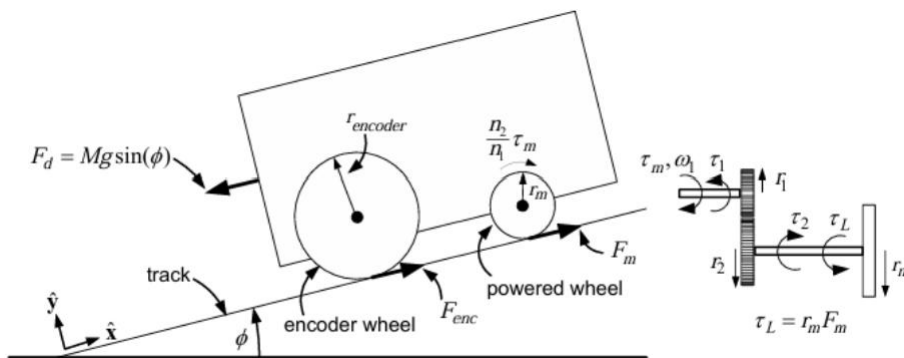


Figure 1: Cart on a inclined floor, powered wheel gears

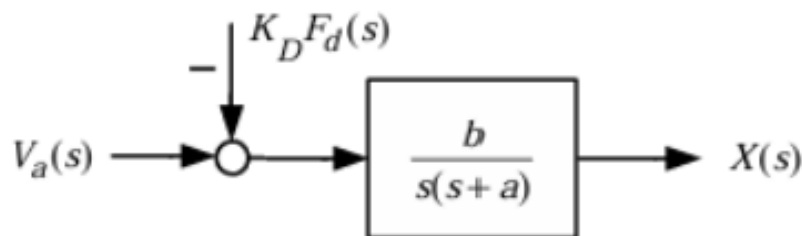


Figure 2: Simplified transfer function of the cart and track system.

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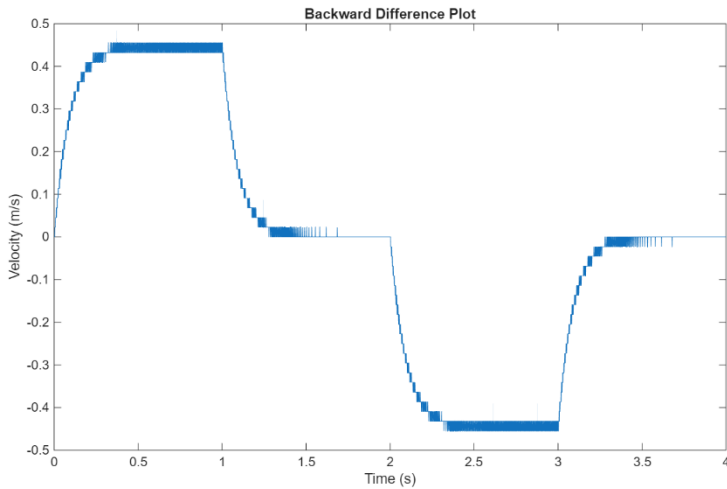


Figure 4: Backward difference plot with the simulation-based analysis

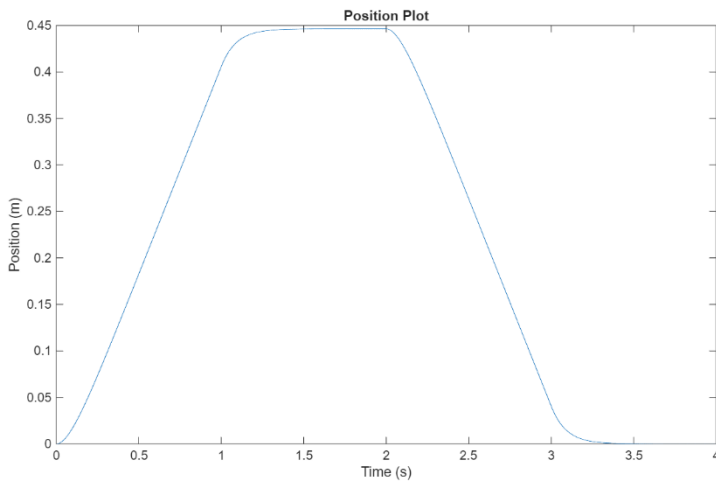


Figure 5: Position plot with the simulation-based analysis

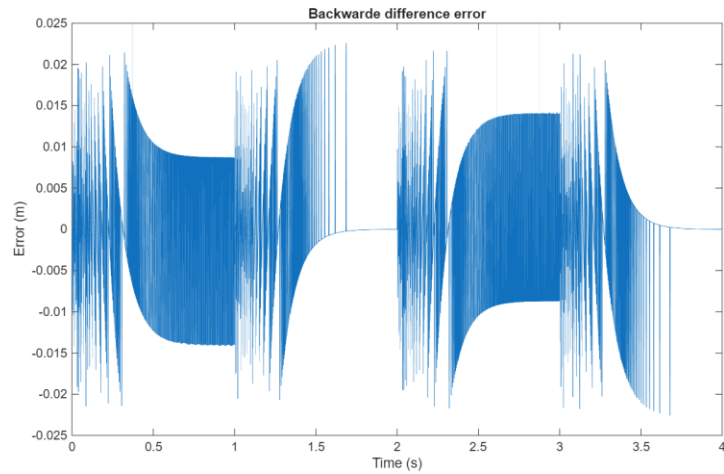


Figure 6: Backward difference error plot of simulated-based analysis

APPENDIX (B)

MATLAB CODE

```

1      %parameters
2      K_T = 7.67*10^-3; %Nm/A
3      R = 2.6; %ohms
4      L = 0.18; %mH
5      M_c = 0.57; %kg
6      M_w = 0.37; %kg
7      r_m = 6.35*10^-3; %m
8      J_1 = 3.9*10^-7; %kgm^2
9      J_2 = 0; %kgm^2
10     J_enc = 0; %kgm^2
11     f_1 = 0; %N*m/rad/sec
12     f_2 = 0; %N*m/rad/sec
13     f_enc = 0; %N*m/rad/sec
14     r_g = 3.71;
15     g = 9.81; %m/s^2
16     N_enc = 4*1024; %counts/rev
17     K_enc = 2.275*10^-5; %meters/count
18     V_max = 5; %V
19     T = 0.001; %s
20
21     %Part A
22     K_b = K_T; %Nm/A
23     M = M_c; %kg
24     J = J_1*r_g^2 + J_2;
25     f = f_1*r_g^2 + f_2;
26
27     a = ((R*f/r_m^2) + K_b*K_T*(r_g/r_m)^2) / (R*J/r_m^2 + R*M);
28     b = (K_T*r_g^2/r_m) / (R*J/(r_m^2) + R*M);
29
30     %Part B
31     phi = 0; %angle
32     K_D = R / (K_T*r_g/r_m);

```

```

34 %Part C
35 load('open_loop_cart.mat')
36 t_model = cart_data(1,:);
37 v_bd_model = cart_data(5,:);
38 v_bd_model = round(v_bd_model,4);
39 dx_model2 = cart_data(3,:);
40 volt_model = cart_data(6,:);
41 position_model = cart_data(4,:);
42
43 figure;
44 plot(t_model,v_bd_model);
45 title("Backward Difference Plot")
46 xlabel("Time (s)")
47 ylabel("Velocity (m/s)")
48
49 figure;
50 plot(t_model,position_model);
51 title("Position Plot")
52 xlabel("Time (s)")
53 ylabel("Position (m)")
54
55 figure;
56 v = out.dx.data;
57 v = v';
58 error = v_bd_model - v;
59 plot(t_model, error);
60 title("Backward difference error")
61 xlabel("Time (s)");
62 ylabel("Error (m)");
63
64 %Part D
65 v_0 = max(v_bd_model);
66
67 [~, idx] = min(abs(v_bd_model - v_0*0.632));
68 closValue = v_bd_model(idx);
69 T_m = t_model(find(v_bd_model == closValue));
70
71 a_model = 1/T_m(1);
72 b_model = v_0*a_model/(max(volt_model));
73
74 %Part E
75 load('Parameter_Identification_Exp_data.mat');
76 t_meas = cart_open_lab01(1,:);
77 x_meas = cart_open_lab01(2,:);
78 dx_meas = cart_open_lab01(3,:);
79 pos_meas = cart_open_lab01(4,:);
80 v_bd_meas = cart_open_lab01(5,:);
81 v_bd_meas = round(v_bd_meas,4);
82
83 v_0 = max(v_bd_meas);
84
85 [~, idx] = min(abs(v_bd_meas - v_0*0.632));
86 closValue = v_bd_meas(idx);
87 T_m = t_meas(find(v_bd_meas == closValue));
88
89 a_meas = 1/T_m(1);
90 b_meas = v_0*a_meas/(max(volt_model));

```